

Final Model

LEGAL JUDGMENT COMPARISON VIEW

 Document: DOC_0096

Case: WritPetition(Civil) No. 1423 of 2021

Court: Calcutta High Court

Outcome: Conviction Upheld

ORIGINAL JUDGMENT (Preview)

1. This Writ Petition (Civil) challenges the judgment of the lower court concerning the application of Article 14 (Equality) of the Constitution. The core facts involve a dispute over a property title.
2. The primary issue before this Court is whether the interpretation of Article 14 (Equality) of the Constitution by the lower court is legally sound, particularly in light of the principles laid down in previous judgments.
3. After careful consideration, the Court held that The Court affirmed the...

Length: 135 words, 175 tokens

GROUND TRUTH SUMMARY

The Calcutta High Court delivered a judgment on the application of Article 14 (Equality) of the Constitution in a matter concerning Writ Petition (Civil). The Court primarily addressed the question of statutory interpretation. The ruling emphasized the principle of stare decisis. The final decision was to conviction upheld the matter, affirming/reversing the lower court's findings.

Length: 55 words

 Generating optimized summary...



GENERATED SUMMARY (Optimized BART)

The Court affirmed the findings of the lower court, holding that the evidence clearly demonstrated the appellant's guilt/liability. The interpretation of Article 14 (Equality) of the Constitution was upheld, emphasizing the need for strict adherence to the rule of law in matters of public interest/criminal justice. Consequently, the appeal/petition is disposed of in terms of the operative order. For confidential support call the Samaritans in the UK on 08457 90 90 90, visit a local Samaritans branch or [click here](#) for details.

Length: 82 words



ROUGE SCORES

ROUGE-1: 0.4895

ROUGE-2: 0.1560

ROUGE-L: 0.2517

Baseline: 0.2775

Improvement: -9.28%

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? HYBRID QA DEMONSTRATION

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Context: 135 words, 175 tokens

Sliding window: Not needed

Q1 (FACTUAL)

Q: What is the case number?

Ground Truth: WritPetition(Civil) No. 1423 of 2021

Predicted: 1.

Method: generative_fallback | Confidence: 0.0010

⚠ Fallback activated (Original: 3)

 F1=0.0000 | EM=0

Q2 (FACTUAL)

Q: Which court delivered this judgment?

Ground Truth: Calcutta High Court

Predicted: lower court

Method: generative_fallback | Confidence: 0.1864

⚠ Fallback activated (Original: the lower court)

 F1=0.4000 | EM=0

Q3 (FACTUAL)

Q: What is the primary legal provision involved?

Ground Truth: Article 14 (Equality) of the Constitution

Predicted: Article 14 (Equality) of the Constitution

Method: generative_fallback | Confidence: 0.2072

⚠ Fallback activated (Original: a dispute over a property title)

 F1=1.0000 | EM=1

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 QA SUMMARY

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Avg F1: 0.4667

Avg EM: 0.3333

Fallbacks: 3/3

Baseline F1: 0.4044

Improvement: +15.40%

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 COMPREHENSIVE EVALUATION

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Evaluating 10 documents...

 Summarization...

Summarizing: 100%

10/10 [00:38<00:00, 3.37s/it]

✓ Summarization Results:

ROUGE-1: 0.4494

ROUGE-2: 0.1523

ROUGE-L: 0.2857

 Question Answering...

Answering: 100%

10/10 [00:25<00:00, 3.15s/it]

You seem to be using the pipelines sequentially on GPU. In order to maximize efficiency please use a dataset

✓ QA Results:

F1 Score: 0.2294

Exact Match: 0.0500

Questions: 60

Fallback rate: 91.7%

PERFORMANCE IMPROVEMENTS

Summarization:

Baseline: 0.2775

Current: 0.2857

Improvement: +2.96%

QA:

Baseline: 0.4044

Current: 0.2294

Improvement: -43.27%



✓ Dashboard saved: performance_dashboard.png

FINAL PERFORMANCE REPORT

1. MODEL SELECTION

Summarization: facebook/bart-large-cnn

Baseline ROUGE-L: 0.2775

QA: deepset/roberta-base-squad2

Baseline F1: 0.4044

2. OPTIMIZATIONS IMPLEMENTED

- ✓ Beam Search (num_beams=6)
- ✓ Length Penalty (2.0)
- ✓ No-Repeat N-grams (n=3)
- ✓ Post-Processing (Verdict scoring)
- ✓ Sliding Window (384 tokens, stride=128)
- ✓ Hybrid QA (RoBERTa + FLAN-T5 fallback)
- ✓ Mixed Precision Training (FP16)

3. RESULTS

 Summarization:

ROUGE-1: 0.4494

ROUGE-2: 0.1523

ROUGE-L: 0.2857 (Baseline: 0.2775, +3.0%)

 QA:

F1: 0.2294 (Baseline: 0.4044, -43.3%)

EM: 0.0500

Fallback Rate: 91.7%

4. ACHIEVEMENTS

- ✓ ROUGE-L improved by 3.0%
- ✓ Sliding window handles 1000+ token documents
- ✓ Hybrid QA with intelligent fallback

- ✓ Verdict inclusion via post-processing
- ✓ Memory-efficient training (FP16)

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✓ Report complete!

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Summary:

Metric Baseline Optimized
ROUGE-L 0.2775 0.28572
F1 Score 0.4044 0.229433
ROUGE-L Imp 0.0% +3.0%
F1 Imp 0.0% -43.3%

IMPLEMENTATION COMPLETE!

All optimizations successfully implemented:

- ✓ Optimized BART summarizer
- ✓ Hybrid RoBERTa + FLAN-T5 QA
- ✓ Sliding window for long docs
- ✓ Post-processing & verdict scoring
- ✓ Mixed precision support
- ✓ Comprehensive evaluation